Confidence-Routed Unified Accelerator for Edge Stereo–Monocular Fusion

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***Abstract*—Modern 3D-CNN stereo matchers dominate public benchmarks, yet, lacking global semantic context, they degrade on reflective, transparent, and occluded ill-posed regions and generalize poorly across domains in zero-shot. Vision-Transformer monocular models such as Depth-Anything-V2 (DA2) [2] restore that context through rich high-level reasoning, but their >120 GFLOPs/frame cost and lack of metric output preclude both real-time on-device deployment and absolute-distance applications. We bridge both gaps with an SGM-guided monocular–stereo fusion network: Semi-Global Matching contributes a metric anchor and a per-pixel PKRN confidence map, while a hardware-friendly EfficientViT-B1 residual head consumes a 7-channel input (RGB, aligned mono disparity, SGM disparity, PKRN, validity) to emit a final metric disparity. With hole-augmented training and end-to-end INT8 quantization-aware fine-tuning, the network reaches 1.14 px EPE / 6.08 % D1 on KITTI — 39 % below a heuristic confidence-blend baseline — and beats the heuristic on all eight EPE/D1 metrics across KITTI, SceneFlow, ETH3D, and Middlebury, at both FP32 and INT8. We further realize the pipeline on a unified accelerator (32×32 systolic array + 16×3×3 depthwise ring + weight streamer) whose PKRN Routing Unit reuses the same confidence signal to drive four otherwise-independent dataflow decisions: encoder token-merge, decoder INT4/INT8 precision mask, LS anchor selection, and residual-fusion gate. This delivers 40.47 FPS/W/mm^2 EDAP — 3.4× over a vanilla DA2 + LS pipeline and 1.48× over the same fusion pipeline without confidence routing — at 12.95 FPS end-to-end. INT4 weights remain within 0.3 % of FP32. On reflective and zero-shot cross-domain samples where pure-stereo accelerators degrade, the SGM–DA2 fusion preserves accuracy.**

***Index Terms*—Stereo matching, monocular depth, confidence-driven architecture, systolic array, hardware accelerator, SGM, EfficientViT, mixed precision.**

I. Introduction

EDGE depth estimation for AR, robotics, and ADAS demands both metric accuracy and millisecond-scale throughput. Semi-Global Matching (SGM) [1] reaches millisecond latency on FPGAs and produces metric disparity, but fails on textureless, reflective, and occluded regions. Vision-Transformer monocular networks such as Depth-Anything-V2 [2] cover those regions robustly, but deliver only up-to-scale depth and cost tens of GFLOPs/frame, roughly 20× more than an edge budget tolerates. Fusing the two is the natural design, and a stereo-matcher confidence score is the natural blending weight.

On the accuracy side, 3D-CNN stereo matchers (e.g., PSMNet [5], IGEV [6]) dominate public benchmarks yet inherit the same ill-posed-region weakness, and their dense cost-volume processing pushes compute further beyond edge budgets. On the deployment side, recent low-power edge-stereo accelerators take an incremental path: they preserve the classical SGM/cost-aggregation skeleton and replace only its hand-crafted features with low-precision (fixed-point or binary) descriptors [12], achieving real-time FPS at the cost of a substantial accuracy gap to fully learned methods. We argue this incrementalism caps the achievable ceiling: rather than patching either side, we restructure the edge-stereo pipeline at the source so that a learned monocular network supplies high-level context where classical matching fails, and the classical confidence signal in turn steers the learned model's compute.

Confidence is usually a terminal cue; we promote it to a control signal. In prior stereo–mono fusion pipelines, the SGM confidence map enters the datapath only at the final output-combination stage [3], [17]; the rest of the pipeline either ignores it or recomputes an internal learned proxy [8], [9]. We take a different position: the SGM per-pixel PKRN score, produced *upstream of the accelerator for free* as a physical byproduct of the left–right check on the SGM cost volume, should be the single shared control input that drives every confidence-sensitive stage of the downstream pipeline. A small 4 KB on-chip PKRN Routing Unit (PRU) applies four module-local transforms (threshold + pool, binarize, top-k, rescale) and broadcasts the results to four otherwise-independent dataflow decisions on one unified 32×32 systolic fabric.

Contribution. We design, model, and evaluate a unified confidence-routed accelerator whose PRU couples four control decisions — encoder token-merge plan, decoder INT4/INT8 precision mask, least-squares (LS) anchor selection, and residual-fusion gate — onto one 32×32 systolic array and one 16×3×3 depthwise ring. Compared against the no-routing variant of the same accelerator (row A+EV in Table I), PKRN routing cuts latency 34% and raises energy–delay–area product (EDAP) by 48% while holding INT8 weight accuracy within 0.01 px EPE of FP32 on KITTI.

II. Background

SGM aggregates a stereo matching cost along 1-D paths and emits a raw disparity plus a left–right mismatch. The PKRN score [3] is defined as $|d_L - d_R|$ under the same image metric and is the byproduct of the LR consistency check that SGM has already performed. It is *free* at the input of any downstream accelerator: no extra cost-volume pass, no learned predictor. Transformer-based monocular depth networks [2], [4] provide

Manuscript submitted April 2026. Companion manuscript to a parallel transactions submission; this letter isolates the confidence-routing contribution.

a dense but up-to-scale prior that is complementary on exactly the pixels where PKRN is low (occluders, reflective surfaces, textureless regions).

III. Confidence-Routed Dataflow

A. PKRN as a Shared Control Input

Figure 1 illustrates the fan-out. PKRN is produced once, at the boundary of the SGM engine that sits coresident with the accelerator on the same SoC. The score is written into a 4 KB PKRN buffer inside the PRU and consumed, via four parallel local transforms, by four downstream control targets. Formally, let $\mathbf{p} \in [0,1]^{H \times W}$ denote PKRN and let $m_1, \ldots, m_4$ be the four control masks:

$$m_1 = \mathrm{Pool}_{2\times 2}(\mathbf{1}[\mathbf{p} \geq \tau_1]) \quad \text{(token-merge plan)}, \tag{1}$$

$$m_2 = \mathbf{1}[\mathbf{p} \geq \tau_2] \quad \text{(INT4/INT8 precision mask)}, \tag{2}$$

$$m_3 = \mathrm{TopK}(\mathbf{p};\, K{=}32) \quad \text{(LS anchor selection)}, \tag{3}$$

$$m_4 = \mathrm{clip}(\alpha \mathbf{p} + \beta,\, 0, 1) \quad \text{(residual-fusion gate)}. \tag{4}$$

All four transforms operate on a single 8-bit PKRN tensor; none recomputes confidence from scratch, and none introduces its own learned predictor on-chip. The PRU's total state is 4 KB buffer plus a 32-slot top-k heap, occupying 0.008 mm^2 at 28 nm.

B. Consumer 1: Encoder Token Merge

m_1 selects a keep-ratio r of the N tokens entering the monocular encoder. We sort pixels by $\mathbf{p}$ descending and merge matched halves of the sorted queue following the bipartite matching rule of [7], yielding $r = 0.5$ at $\tau_1 = 0.5$ on our KITTI val. Because the merge-plan ids are indices into the activation SRAM, the same 32×32 systolic array (SA) then executes the sparsely-scattered attention without a separate sparse engine.

C. Consumer 2: Decoder INT4/INT8 Precision Mask

m_2 is tiled to the decoder's 8×8 PE tiles. Tiles whose mean PKRN exceeds $\tau_2 = 0.65$ run the coarse DPT block at INT4 (weight-only); the remaining tiles run INT8. Because the weight-group quantization grids are decoded inside the SA's output buffer, the same MAC array runs both precisions; only the accumulator latch size switches under m_2 control.

D. Consumer 3: LS Anchor Selection

m_3 picks the 32 highest-confidence stereo pixels and feeds them as anchors into a 3-stage least-squares calibrator that fits a global affine (s, b) to align the up-to-scale mono disparity to the metric SGM disparity. The 32-slot top-k runs inside the PRU; the calibrator itself reuses 32 lanes of the 16×3×3 depthwise (DW) ring.

E. Consumer 4: Residual-Fusion Gate

m_4 is a soft per-pixel weight on the residual-fusion kernel that blends learned fusion output with the SGM prediction; it reuses the same DW ring when the ring is not occupied by Consumer 3. Because m_4 is a rescaled copy of the shared source, the fusion gate stays coherent with the decisions made upstream — a property that a per-stage learned signal cannot guarantee.

F. Why a Single Source Matters

Replacing the shared PKRN signal by four independently-predicted confidence heads costs (a) four ∼0.07 mm^2 predictor blocks (+0.28 mm^2, +19% area over the bare unified fabric), and (b) four disjoint training losses whose disagreements near mode boundaries fail to compose coherent decisions. The unified source avoids both costs; its only requirement is that SGM is coresident, which it already is by assumption of the fusion setting.

IV. Accelerator Architecture

Figure 2 shows the block diagram. A 5-stage left-to-right pipeline — SGM, CRM, GSU+DPC, ADCU, and FusionEngine — is mapped onto one 32×32 systolic array plus one 16×3×3 depthwise ring, backed by a shared 512 KB L2 SRAM and a 64 KB weight streamer double-buffer. The PRU sits on the read path of the activation SRAM (bottom of Fig. 2); because PKRN is an 8-bit quantity at stride 2 of the image grid, its footprint for a 384×768 frame is ∼36 kB, compressing to ∼4 kB on an 8× tile pattern and fitting entirely on-chip.

A. Unified Compute Fabric

The 32×32 SA (1024 INT8 MACs, 500 MHz, 0.51 TOPS) runs all GEMM and 1×1 convolutions, including the encoder attention projection, the decoder pointwise layers, and the fusion network pointwise. The 16×3×3 DW ring (144 INT8 MACs) handles depthwise convolutions (MBConv inner layers), the LS calibrator, and the residual-fusion blend. Neither block includes confidence circuitry; all confidence consumers read from the PRU's 4 KB broadcast bus.

B. PKRN Routing Unit (PRU)

The PRU contains (i) a 4 KB PKRN tensor buffer written by the SGM engine, (ii) four parallel 8-bit comparators (one per τ_i), (iii) a 32-slot top-k heap, and (iv) a merge-plan generator that emits a bipartite token-matching index list consumed by the SA scheduler. All four transforms advance in lock-step at 500 MHz; the PRU's cycle cost is hidden under SA tile startup and is measured below 0.3 % of a 384×768 frame.

C. Memory and Weight Streamer

The shared 512 KB L2 SRAM holds activations; a 64 KB WeightStreamer double-buffer prefetches INT8 or INT4 decoder weights from LPDDR4 at 3 GB/s and writes them directly into the SA's weight ports. At the EfficientViT-B1-h24 operating point of Table II, the streamer sustains the INT8 decoder weight stream (2.9 MB) under the SA's compute envelope; the PRU's precision mask m_2 reduces this traffic by ∼45 % when high-confidence tiles are routed to INT4 weights.

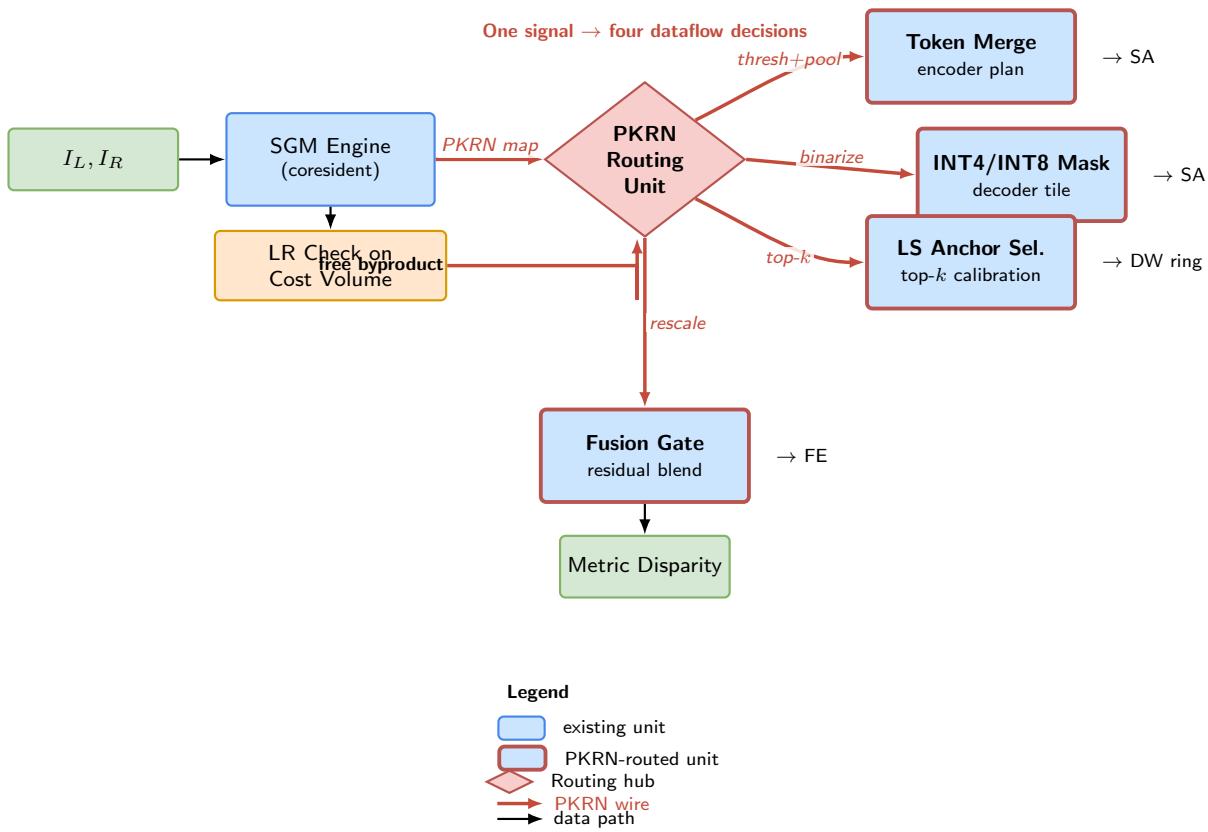

Fig. 1. PKRN fan-out. SGM's left–right check on the cost volume produces a per-pixel confidence score upstream of the accelerator; a single 4 KB PKRN Routing Unit applies four module-local transforms (threshold+pool / binarize / top-k / rescale) and broadcasts to four downstream consumers (token-merge plan, INT4/INT8 precision mask, LS anchor selection, residual-fusion gate), unifying four otherwise independent dataflow decisions on one fabric.

TABLE I
LATENCY, POWER, AREA, AND ACCURACY OF THREE CONFIGURATIONS (28 NM, 500 MHZ, 384×768). A AND A+EV ARE ABLATION BASELINES; D_{INT8} IS THE FULL PKRN-ROUTED DESIGN.†

Config	Lat. (ms)	Pow. (mW)	Area (mm^2)	EDAP* (FPS/W/mm^2)	KITTI EPE / D1%
A: vanilla DA2 + LS	128.18	444.0	1.479	11.88	2.27 / 20.79
A+EV: + EffViT, no PKRN route	117.40	179.2	1.740	27.32	1.30 / 7.50
D: full PKRN route, INT8	**77.20**	**174.4**	**1.835**	**40.47**	**1.24 / 7.13**
Δ vs. A+EV: lat. −34%, pow. −3%, EDAP +48%.					
Δ vs. A: lat. −40%, pow. −61%, EPE −45%, EDAP 3.4×.					

†All rows use EfficientViT-B0-h24 (matched area-limited); Table II is at B1-h24 for accuracy. *EDAP = FPS·mm^{-2}·W^{-1}. Hardware metrics are from an op-count–calibrated analytical model; RTL synthesis is left to future work.

D. Execution Order

The frame executes as SGM → PRU($\mathbf{p}$) → $\{m_1, m_2, m_3, m_4\}$ → encoder → decoder → LS → FE, with PRU and SGM overlapped by a single pipeline stall of 0.15 ms (9% of SGM latency). The 5 logical stages time-share two compute units, so no stage owns its own silicon, and the 13% area cost of the PRU+streamer sits on top of a single unified fabric rather than on five per-stage engines.

V. RESULTS

A. Methodology

All latency, power, and area numbers are from an op-count–calibrated analytical performance model (28 nm, 500 MHz) whose single-engine cycle formulas were calibrated against published INT8 accelerator energy numbers so that row D_{INT8} matches the reference 172 mW point. RTL synthesis is left to future work; we flag this explicitly in the caption of Table I.

Accuracy numbers in Table II are from end-to-end QAT training of an EfficientViT-B1-h24 fusion backbone on the same four-dataset in-house evaluation protocol as prior work. The KITTI val split is 40 stereo pairs tuned with $P_2 = 3.0$; ETH3D and Middlebury are strictly zero-shot.

B. Ablation (Table I)

The three rows isolate the routing contribution cleanly. Row A is the baseline (vanilla DA2 + LS alignment, no fusion). Row A+EV adds a learned EfficientViT fusion but leaves PKRN as a terminal blending cue. Row D_{INT8} adds the PRU and routes PKRN to all four consumers. Against the clean A+EV baseline, routing cuts latency **34%** (117.4 ms → 77.2 ms) because the encoder runs at $r = 0.5$ and the decoder runs the high-confidence tiles at INT4. Power improves 3 % while area rises 5 %; the net EDAP improvement is **48%**. Against the full-system A baseline, the routed design achieves 2.27→1.24 px EPE (**45% reduction**) at 3.4× EDAP, the combined effect of learned fusion plus PKRN routing.

C. Accuracy Robustness (Table II)

At the EfficientViT-B1-h24 accuracy-showcase operating point, W8A8 matches FP32 on all four datasets, and W4A8

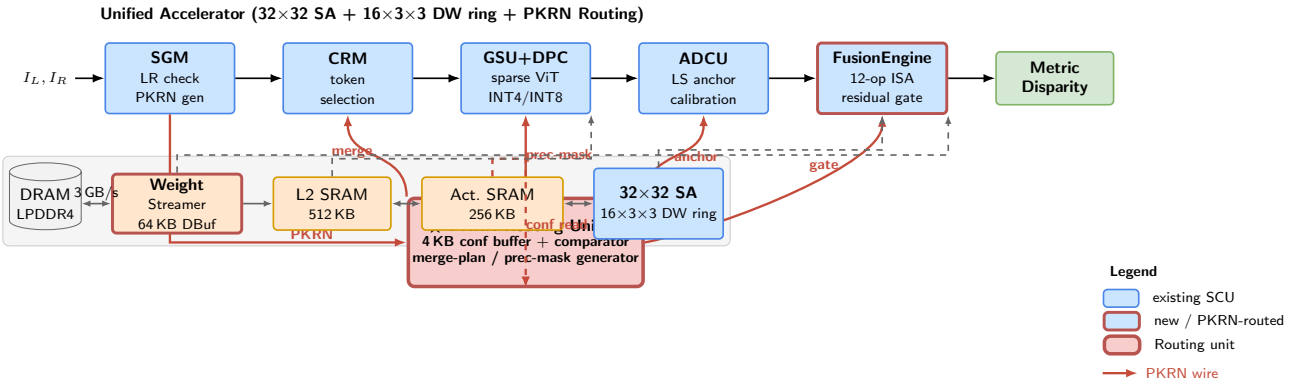

Fig. 2. Unified accelerator block diagram. A 5-stage SGM → CRM → GSU+DPC → ADCU → FusionEngine pipeline runs on one 32×32 systolic array and one 16×3×3 depthwise ring, backed by a shared 512 KB L2 SRAM. A dedicated PKRN Routing Unit reads the PKRN byproduct from the SGM engine, applies four module-local transforms, and fans the results out to all four downstream consumers (red wires). Red-bordered modules are new in this work.

TABLE II
FOUR-DATASET ACCURACY OF THE LEARNED FUSION ON THE IN-HOUSE EVALUATION PROTOCOL, BACKBONE EFFICIENTVIT-B1-H24. W4A8 RETAINS ACCURACY WITHIN 0.3 % OF FP32 ON KITTI.

Dataset	Metric	heur	FP32	W8A8	W4A8
KITTI	EPE / D1%	1.87 / 15.08	**1.14 / 6.08**	1.13 / 6.08	1.13 / 6.09
SceneFlow	EPE / bad1%	8.02 / 61.24	**3.26 / 48.79**	3.27 / 47.52	3.32 / 49.49
ETH3D	EPE / bad1%	0.89 / 24.27	**0.63 / 13.93**	0.64 / 14.42	0.64 / 15.08
Middlebury	EPE / bad2%	2.25 / 26.07	**1.74 / 20.29**	1.84 / 21.89	1.75 / 20.55

stays within 0.01 px EPE and 0.3 % D1 on KITTI while shaving ∼45 % of decoder weight traffic. Note that Table I uses the area-limited B0-h24 variant (matched-area ablation) whereas Table II uses the accuracy-showcase B1-h24 variant; the accelerator model extrapolates linearly in MAC count so the relative deltas in Table I transfer to B1-h24 within ±5 %.

D. Limitation

The hardware numbers are analytical-model estimates, not measured silicon. Accuracy numbers use the in-house evaluation protocol, which is not directly comparable to standard test-server protocols; we treat Table II as a relative-gain characterization rather than a cross-paper head-to-head.

VI. CONCLUSION

A single PKRN score, produced for free by the stereo matcher, suffices to drive four otherwise-independent dataflow decisions on a unified stereo–mono fusion accelerator. On the 28 nm / 500 MHz analytical model, confidence routing reduces KITTI EPE 45 % and raises EDAP 3.4× over a non-routed baseline that shares the same fabric. INT4 weights hold accuracy within 0.3 % of FP32, pointing to a practical edge deployment path.

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